Rangaswamy A

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SUMMARY

Aspiring AI/ML Engineer with hands-on experience building practical AI solutions. Passionate about applying machine learning and artificial intelligence to solve real-world problems and develop practical applications.

SKILLS

- **Programming Languages**: Python, SQL
- **Machine Learning:** Supervised & Unsupervised Learning, Ensemble Learning, Model Evaluation, Regularization (Lasso, Ridge), Overfitting & Underfitting
- **Machine Learning Algorithms:** Linear Regression, Logistic Regression, Decision Trees, SVM, KNN, Naive Bayes, K-Means, DBSCAN, PCA, Hierarchical, Isolation Forest, LOF
- **Deep Learning:** ANN, FNN, CNN, RNN, LSTM, Transformers
- **Generative AI:** LLMs, NLP, RAG, AI Agents, OpenAI API, GAN
- **Data Analysis & Visualization:** NumPy, Pandas, Matplotlib, Seaborn
- **Web Technologies:** HTML, CSS
- **Frameworks & Tools:** Scikit-learn, PyTorch, Flask, FastAPI, Streamlit
- **Developer Tools:** Git, GitHub, Docker, CI/CD

EDUCATION

- B.E. in Artificial Intelligence & Machine Learning | ACET College | CGPA: 7.63 | Currently Pursuing 7th Semester
- XII (DPUE) |Govt. Ex-Municipal PU College *86.5%| 2023*

ACADEMIC PROJECTS

AI-Based Smart Attendance System — Individual Project

- Developed an AI-powered attendance system using face recognition and voice recognition to automate student attendance with separate Teacher and Student portals.
- Implemented Computer Vision and Speaker Recognition pipelines using Scikit-learn (SVM), dlib, and Resemblyzer, with Supabase for secure authentication and attendance management.
- Technologies: Python, Streamlit, Scikit-learn, dlib, Resemblyzer, Librosa, Supabase, SQL, Git
- GitHub: https://github.com/RangaA-sml/AI_Attendance_App
- Live Demo: *https://fast-attendance-ai.streamlit.app/*

AI YouTube Video Analyzer — Individual Project

- Developed an AI-powered YouTube Video Analyzer using Agno AI Agent and Groq LLM to generate video summaries, detailed analysis, study notes, key takeaways, quizzes, and timestamp-based insights from YouTube videos.
- Built an interactive Streamlit application with YouTube metadata extraction, AI-powered content analysis, and downloadable PDF/Text reports for improved learning and content understanding.
- Technologies: Python, Streamlit, Agno AI, Groq LLM, yt-dlp, YouTube Tools, ReportLab.
- GitHub: https://github.com/RangaA-sml/AI_YouTube_Video_Analyzer
- Live Demo: https://aiyoutube-video-analyzer.streamlit.app/

AI Real-Time Gym Trainer — Individual Project

- Developed an AI-powered real-time fitness trainer using MediaPipe Pose Estimation to detect exercises, count repetitions, and analyze workout form for Squats, Push-ups, Lunges, Shoulder Press, and Biceps Curls.
- Integrated an LLM-based AI Coach with voice feedback to provide real-time posture corrections and workout guidance, while tracking workout history and performance.
- Technologies: Python, Streamlit, MediaPipe, OpenCV, Llama 3, Groq API, gTTS, SQLite, NumPy, WebRTC
- GitHub: *https://github.com/RangaA-sml/RangaA-sml-AI_Real_Time_Gym_Coach*
- Live Demo: *https://aireal-time-gym-coach.streamlit.app/*

Live Demo Note: Applications are deployed on Streamlit Community Cloud. If an app is inactive, click **"Yes, get this app back up!"** to wake it up; the application will then load normally.

CERTIFICATION

Prime (AI/ML) — Certificate of Completion, Apna College | [View Certificate]